

G Glow

An 8-voice additive synthesizer in the spirit of the first micro-computer digital synths of 1980-81.



Overview

G Glow builds every sound from harmonic partials. You draw the level of each partial on a green phosphor screen, and the synth turns that drawing into a waveform. It is modelled on the early home-computer digital synths, where a pair of oscillator cards sat inside the computer and you drew the sound on the monitor.

Each voice has two paths, A and B. A path is an additive oscillator (16, 32 or 64 partials) plus a noise generator, with its own envelope, level and pan. The two oscillators can modulate each other (X-MOD), two LFOs add movement, and a stereo dual delay finishes the sound.

Each oscillator can run MODERN - clean and band-limited, with no aliasing - or VINTAGE, which recreates the old card: an 8-bit, 256-byte wavetable read with no

smoothing, an 8-bit volume stage that makes the tails step down, and a 31.25 kHz output clock. VINTAGE aliases at the top on purpose; that grit is the sound.

Reach for it for

- Glassy, bell-like and chime sounds built partial by partial
- Lo-fi 8-bit basses, leads and arcade percussion (VINTAGE with LFSR noise)
- Organ, string, flute and choir-like tones from drawn spectra
- Slow-moving pads and drones with stereo LFO panning and the dual delay
- Quick sound design: RANDOM makes a new, usable partial set in one click

Playing it

G Glow is an instrument: it has no audio input and makes its sound from MIDI.

1. Insert G Glow on a track (as the first plug-in, or after a MIDI effect).
2. Arm the track for recording and set its input to your MIDI keyboard (or draw notes in a MIDI item).
3. Play. The MIDI light in the header flashes on each note, and the patch display shows how many of the 8 voices are sounding.
4. If you use a multi-channel MIDI setup, set CHANNEL in the MASTER section to the channel you want; OMNI listens to all of them.

G Glow responds to pitch bend (range set by BEND), the mod wheel (CC1 adds to LFO 1's amount, up to 100%), the sustain pedal (CC64) and all-notes-off (CC120 and CC123). Incoming MIDI is passed on down the chain.

Controls

The header

< and >	Step to the previous or next factory patch.
Patch display	Shows the patch number and name, the voice mode, how many of the 8 voices are playing and SUS while the sustain pedal is down. Click it to open the patch list.
PRESETS	Also opens the patch list. A patch sets every sound control and both partial sets; the MIDI channel, DRIFT, VOLUME,

	the STEREO controls and the output stage are left as they are.
MIDI	Lights when a note arrives.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
PANIC	Stops every sounding note at once. Changing the voice mode does the same.

Oscillator A and Oscillator B

Each oscillator has a screen showing its partials as bars, from the fundamental (1) on the left upwards. The small window at the top right of the screen shows the single wave cycle the partials make.

The screen	Left-drag to draw partial levels; drag across the screen to draw a whole curve in one go. Right-drag to erase. The partial sets are saved with the project.
MODERN / VINTAGE	MODERN: a clean oscillator that leaves out any partials that would go past the top of the audio range, so it never aliases. VINTAGE: the old 8-bit card - a 256-byte wavetable with no smoothing, pitch steps from the old 16-bit counter, an 8-bit volume stage after the envelope (stepped tails) and a 31.25 kHz output clock. It aliases at the top, and the screen shows levels in 16 steps, like the old monitor.
16 / 32 / 64	How many partials the oscillator uses. Fewer partials give a softer, simpler sound and wider bars to draw on.
RANDOM	Makes a new partial set that is still musical: a falling harmonic series with a little variation, odd partials only, random spikes over a fundamental, one to three formant bumps, or a saw with every few partials missing.

SHAPES

A menu of starting points: Sine, Saw, Square, Triangle, Odd emphasis, Soft decay, Formant, Organ (drawbar-style), Even emphasis, Bright tine, and Clear (all partials off).

Path A and Path B

TRANPOSE	The oscillator's pitch in semitones, -36 to +36.
FINE	Fine tuning, -100 to +100 cents. Opposite settings on A and B give a detuned, chorused width.
PITCH ENV	How far the path's own envelope bends its oscillator's pitch, -48 to +48 semitones at the top of the envelope. Use it with a short decay for thumps, drops and zaps.
LEVEL	The path's volume, 0 - 100%. Path A starts at 80%, Path B at 0%, so a fresh patch only plays A.
PAN	Where the path sits in the stereo field, from L 100 to R 100.
NOISE	Blends from the oscillator to the noise generator: 0% is all oscillator, 100% all noise.
TONE	The brightness of the noise, from about 200 Hz up past the top of the audio range. With WHITE it is a low-pass filter; with LFSR it is the clock rate of the noise.
WHITE / LFSR	WHITE: smooth hiss. LFSR: a 15-bit shift-register noise, the gritty, metallic noise of 80s digital chips - good for snares, hats and arcade effects.

Envelope A and Envelope B

Each path has its own ADSR envelope, which shapes its level and its PITCH ENV. The small screen to the right sketches the envelope's shape.

ATTACK	Time to rise to full level, 0.5 ms to 10 s.
DECAY	Time to fall to the sustain level, 1 ms to 20 s.
SUSTAIN	The level held while the key is down, 0 - 100%.

RELEASE	Time to die away after the key is let go, 1 ms to 20 s.
A CURVE / D CURVE / R CURVE	The shape of each stage, -100 to +100. 0 is a straight line; positive values are exponential and snappy; negative values start slowly. D and R start at +50.

Voice

POLY / MONO / RETRIG	POLY: up to 8 notes at once. MONO: one note, legato - when you play a new note while holding another, the pitch glides and the envelopes carry on. RETRIG: one note, and every note restarts the envelopes.
GLIDE	How long the pitch takes to slide to a new note, 0 - 2000 ms. In POLY each voice glides from the last note it played.
BEND	Pitch-bend range, 0 - 24 semitones.
VELOCITY	How much playing harder makes the sound louder, 0 - 100%. At 0% every note plays at full level.

X-MOD

A > B	Oscillator A modulates the phase of oscillator B (frequency modulation in the FM-synth manner), 0 - 100%. Adds bright, bell-like and metallic overtones to B. A is used as a modulator even if its LEVEL is at 0.
B > A	The other way round: B modulates A.

LFO 1 and LFO 2

WAVE	SINE, TRI, SQUARE or S&H (random steps).
RATE	LFO speed, 0.02 - 30 Hz.
SKEW	Shifts the shape. On SINE and TRI it moves the peak earlier or later; on SQUARE it sets the pulse width. On S&H, settings above 50% make the steps glide into each other.

DEST	What the LFO moves: OFF, PITCH (both paths), PITCH A, PITCH B, PAN (A one way, B the other), LEVEL (tremolo), X-MOD (the depth of both X-MOD knobs) or NOISE (the noise blend).
AMOUNT	How much it moves, 0 - 100%. On pitch the range grows towards the top of the knob, up to 12 semitones at 100%, so small settings give fine vibrato. The mod wheel adds to LFO 1's amount.
POLY / GLOBAL	POLY: every note has its own LFO, restarted when the note starts - each note gets the same vibrato or sweep. GLOBAL: one LFO for all notes, running freely. LFO 1 starts on POLY, LFO 2 on GLOBAL.

Dual delay

ON	Switches the delay in.
PING	Ping-pong: the sound goes into the left side only (summed to mono), and CROSS bounces it across to the right. Off: stereo in, stereo out.
SYNC L / SYNC R	The delay time of each side as a note value of the project tempo, from 1/32 to 1/1 (with triplet and dotted values), or OFF (ms) to set it in milliseconds.
TIME L / TIME R	The delay time in milliseconds, 1 - 2000 ms, used when that side's SYNC is OFF. The label is shown in brackets while the side is synced.
FEEDBACK	How much of each side repeats into itself, 0 - 95%.
CROSS	Cross-feedback: how much of each side repeats into the other, 0 - 95%. FEEDBACK and CROSS together are held below the point where the repeats would build up, so the delay always dies away.
LP / HP	Filters inside the feedback loop: LP (500 Hz - 20 kHz) darkens each repeat, HP (20 Hz - 2 kHz) thins it. The repeats get darker or thinner as they go.

MIX

How loud the delay is, 0 - 100%. The dry sound stays at full level.

Drift

A slow random wander, like old analogue tuning. DRIFT sets how much (0 - 100%, starting at 20%). In G Glow it moves the pitch of the whole instrument (up to 20 cents either way at 100%), lets oscillator B wander against A by up to 0.15 semitones, and gives each new note a slightly different tuning (up to 6 cents).

Master

DRIFT	See above.
VOLUME	Output level, -24 to +6 dB.
CHANNEL	The MIDI channel G Glow listens to: OMNI (all) or CH 1 to CH 16.
Meters	Output level, left and right.

IN / OUT strip

The slim strip along the bottom of the panel.

SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). CLIP is on by default; its key flashes red while it is clipping.
MONO	Makes everything below this frequency mono, from 20 to 500 Hz (fully left = OFF). Keeps the low end centred when paths are panned apart or the delay is wide.
WIDTH	Widens the stereo image above the MONO frequency, up to +100% (twice the side signal).

Presets

Each patch is levelled so that patches play at a similar loudness.

1981 Micro Chime	Both oscillators VINTAGE with 16 partials, B an octave up; fast-decaying chime, a touch of vibrato and a ping-pong delay.
Lo-Fi Bit Bass	MONO legato with a short glide; a VINTAGE bass made from the first four partials, a little LFSR noise and a sine an octave below on B.
Spectral Ambient Pad	MODERN with 64 partials, a formant set on A and a soft decay on B, detuned 7 cents either way; slow attack, slow LFO panning and a long delay.
E.Piano Tine	A bright tine on A with a fast decay over a warm sine body on B, a little A > B modulation and a gentle stereo pan.
Digital Sci-Fi Drone	Heavy A > B modulation with S&H on the X-MOD depth, slow panning and a dark, long-feedback delay.
8-Bit Arcade Percussion	A VINTAGE thump that drops in pitch (PITCH ENV) and an LFSR noise snap on B.
Dual-Timbral Poly	A saw on A hard left and a square on B hard right, slightly detuned, with a short delay.
8-Bit Arcade Snare	A short pitched thump on A and a bright LFSR noise body on B.
Glass Organ	VINTAGE drawbar-style partials on A, even partials an octave up on B, organ envelopes and a light vibrato; velocity off.
Vintage Strings	Two detuned VINTAGE saws with a slow swell, per-note vibrato and slow panning.
Hollow Flute	Odd partials an octave up with a little breath noise, per-note vibrato and a short delay.
Bell Tower	A spread, bell-like partial set on A and a high, detuned

	strike on B, long decays and a long delay.
Choir Formant	Two 64-partial formant sets, detuned, with slow swells, vibrato and slow panning.
Square Sync Lead	MONO legato with glide; a bright square on A, modulated by B ($B > A$), with vibrato and delay.
Init	A plain starting point: one MODERN saw on A, Path B silent, no LFOs, no delay.

Tips

- Start from SHAPES, then draw on top: pull a few partials up or down and listen to how the tone changes.
- For a classic 80s digital sound, set both oscillators to VINTAGE and use 16 partials; for a clean, hi-fi version of the same drawing, switch back to MODERN.
- Give Path B a short envelope and a different partial set to add a separate attack or strike to a sustained sound.
- Small amounts of $A > B$ (5 - 15%) brighten a sound without making it obviously metallic.
- If the delay or panned paths make the low end wander, raise MONO in the IN / OUT strip to 100 - 150 Hz.

File: gfx-g-glow.jsfx